

NEF 3001- Applied Project 1

Summer Block Group 1

PROGRESS REPORT

for

SMART LOCKER SYSTEM



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Supervisor: Dr.Assefa K Teshome

Sponsor: A/Prof Thinh Nguyen

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Introduction

This is our group's finished progress report for the product "Food for Dirt". At the start of the project, our group had 2 documents to keep track of our project's progress. They were the UI DB.docx (The user front end app) and the locker system.docx (program written to control the lockers). This document is the edited and the combined version of those 2 documents. Our Group kept track of our meeting records provided at the bottom of this document. Also included in this document are the rough database schema and the design constants we initially began with.

After setting up GitHub to work on the project we realized that when pushing a commit, we could have done the same thing and detailed any changes in the commit, but since we had already started this way, we decided to continue.

User app Git

<https://github.com/fwvu/UI-database>

Locker System program Git

<https://github.com/fwvu/LockerBank>

Viewable current Gantt Chart

https://livevuedu-my.sharepoint.com/:x:/g/personal/bryan_france_live_vu_edu_au/EYFdxqp-vK9Fgk9fSMhT0rsBw5Ci4wPbD9O6c8IxR-2yWA?e=uNWwqa

Viewable original user app progress doc UIDB.docx (moved here and discontinued)

https://livevuedu-my.sharepoint.com/:w:/g/personal/bryan_france_live_vu_edu_au/EQJTHiESYtpBuAAHiTq1y48BnjX_pyiSfCVHSIL_sMtGdQ?e=eeDWeU

Viewable original locker system progress doc LockerSystem.doc (moved here and discontinued)

https://livevuedu-my.sharepoint.com/:w:/g/personal/bryan_france_live_vu_edu_au/EaBxrC5iPtBEsee8cXimHxgBU9ZAzLGQA25RUhi25NUjdg?e=wz92Xr

App Progress Report

Sign up Subsystem.

SIGN UP	
Files Used	
signup_screen.dart	
signup_footer_widget.dart	
signup_form_widget.dart	
signup_controller.dart	
signup_email_password_failure.dart	

Date	Team Member	Notes
07/03/2023	Bryan	Created custom generic form header (requires image, title, and subtitle) Created custom signup form widget that contains all the text fields and the sign-up button. Created text field theme. Created custom signup footer widget for google sign in and login link. Added G logo to image assets folder
09/03/2023	Peter	Added international phone number textbox for the signup bar. Added Button Functionality to go to the login page if user already signed up.
10/03/2023	Peter	Added password obscuring code using the command obscure=true to hide the password, although the button to make the password visible (the eye icon) is not working
28/03/2023	Bryan	Added validation email and username cannot be empty. Added firebase validation for email and password quality (ie: length strength) Added if account creation fails no database record is created. Restored international phone number text field (not require for authentication now)

Login/OTP Screen/Forgot Password Subsystem

LOGIN	
Files Used	
login_footer_widget.dart	
login_form_widget.dart	
login_screen.dart	
otp_Controller.dart	
Forget_password_mail.dart	
Forget_password_btn_widget.dart	
Forget_password_model_bottom_sheet.dart	
otp_screen.dart	

Date	Team Member	Notes
11/03/2023	Peter	Created custom generic form header (requires image, title, and subtitle) Created custom Login form widget that contains all the text fields and the Login button. Created text field theme. Created custom login footer widget for google sign in and sign-in link. Added password obscuring code using the command obscure=true to hide the password, although the button to make the password visible (the eye icon) is not working Added Button Functionality to go to the sign-up page if user already signed up Added Forget Password Button Functionality
12/03/2023	Peter	Added Forget password window to recover/edit new password through email. Added button widgets for the forget password. Added the window for the forget password. Added OTP (One Time Password) screen (Still WIP)
13/03/2023	Peter	Continued working with the OTP (OTP is not getting recognized by the flutter app, so our group tried removing the Intl phone number for now)
15/03/2023	Peter	OTP Is removed for now since we are advised by both our sponsor and advisor (Assefa and Thinh respectively)

Firebase Authentication Subsystem

FIREBASE AUTHENTICATION	
Files Used	
authentication_repository.dart	
profile_controller.dart	
user_repository.dart	
user_model.dart	

Date	Team Member	Notes
13/03/2023	Peter	Created the firebase (Food For Dirt Testbed) window to prepare the firebase and flutter integration Created The authentication repository file for the firebase integration with flutter
14/03/2023	Peter	Created the Profile Controller file to register user creation to the firebase database. Created The user model file
18/03/2023	Peter	First Attempt with firebase rule for specified user can only access their own database from their UserID (Failed, we changed the rules by turning off security entirely)
20/03/2023	Peter	Second Attempt with firebase rule for specified user can only access their own database from their UserID (Failed, the rule is changed to read-only and none of the user can access the specified database)
29/05/2023	Peter	Third Attempt with firebase rule for specified user can only access their own database from their UserID (to be fixed further if required)
4/06/2023	Peter	Fourth and final Attempt with the firebase rule, it has now been setup properly

Dashboard / User Portal Subsystem

DASHBOARD / USER PORTAL	
Files Used	
dashboard.dart	
profile_menu.dart	
edit_profile_screen.dart	
profile_screen.dart	
profile_controller.dart	
user_repository.dart	

Date	Team Member	Notes
15/03/2023	Peter	Added simple dashboard with a welcome text and a sign out button functionality
16/03/2023	Peter	Added basic profile window (To be reworked later down the line)
22/03/2023	Bryan	Altered dashboard. Added working QR code to display on dashboard made from <code>QrImage("Food For Dirt \$email")</code>– when user logged in is checked on authentication repository if a user exists the user email data is passed to the dashboard screen, "Food For Dirt is concatenated with email " Created Profile screen and Edit Profile screen. Profile screens have only placeholder buttons currently. The edit profile button gets to the edit screen and the logout button logs the user out. The other buttons are non-functional. Edit Profile screen uses a Future Builder, checks for a current logged in user, then populates the text fields with the current user's data. The update profile button is currently non-functional. Added functionality to update profile screen. Created the <code>updateUser</code> function to <code>profile_controller.dart</code> which takes a snapshot of the user data on the edit profile screen. Created the <code>updateUserRecord</code> in the <code>user_repository.dart</code> which passes data to firebase. Tested and functional
28/03/2023	Bryan	altered dashboard. Change the display to say "Welcome fullName." Changed QR code to use "Food For Dirt concatenated with user account ID"
20/04/2023	Bryan	Changed QRcode from "Food For Dirt \$userID" to "Food_For_Dirt \$userID". For more efficient use with locker bank

Miscellaneous Features

MISCELLANEOUS	
Files Used	
theme.dart	
elevated_button_theme.dart	
outlined_button_theme.dart	
text_field_theme.dart	
text_theme.dart	
colors.dart	
image_strings.dart	
Sizes.dart	
text_strings.dart	
fade_in_model.dart	
fade_in_controller.dart	
animation_design.dart	
splash_screen.dart	
welcome_screen.dart	

Date	Team Member	Notes
01/03/2023	Bryan	Folder Structure: Created multiple folders to break up the code. This folder structure should allow us to keep various parts of the code in different areas. This will make it easy to alter and manage code more efficiently. I also created a package for the authentication section of the code to keep it separated from the main app.
01/03/2023	Bryan	Constants: Created files for Color, Image strings, Sizes, Text strings. Making it easier to change 1 thing across the whole app. Created assets folder for all images and other assets
04/03/2023	Bryan	Theme Setup: Basic setup for themes light and dark mode. More will be added during the project. theme.dart file. Text theme done; brightness set. Elevated button theme created.
05/03/2023	Bryan	Outlined button theme created. light and dark text theme created.

04/03/2023	Bryan	Splash Screen: Setup animated splash screen using GetX. Created splash screen, splash screen controller, now 4 widgets animate into the main area and wait for 5 seconds before moving to next screen.
06/03/2023	Bryan	Converted the splash screen animation controller and made use of the custom controller widget.
05/03/2023	Bryan	Welcome Screen: Created Welcome Screen created elevated_theme_button.dart and outlined_button_theme.dart added to the theme.dart file. Added light and dark text to text_theme.dart and added to the theme.dart
06/03/2023	Bryan	Wrapped the welcome screen in a custom animation widget – had to convert it to a STACK widget to be able to animate. had to add all positions as 0 (except –100 for scroll up) so all the contained widgets did not reposition.
06/03/2023	Bryan	Custom Animation Widget: Created a custom animation widget that can now be added to animate any widget.
14/03/2023	Peter	Added Picture (img): Added picture for OTP screen (Could be removed since OTP is not working) Added Text Strings: Added Text Strings for both Forgot password and OTP screen (Still in consideration)

Locker Bank Progress Report

QR Reader Subsystem

QR READER	
Files Used	
Lb_tk_qr1_scanner.py	

Date	Team Member	Notes
20/04/2023	Bryan	Using openCV2 created a loop that will read the camera image. OpenCV2 had a built in QR decoder. setup the code in the loop to read the camera image and target any QR images and output the decoded QR into terminal. Tested successfully.
20/04/2023	Bryan	Implementing tkinter. Progress moved to SCREEN OUTPUT section. Only major changes need to be added to this section.

Locker Selection Subsystem

LOCKER SELECTION	
Files Used	
Lockertester.py	
Lockercheck.py	
Lockerreset.py	
Lockers.dat	
Lbtkinterpractice.py	
Lbtkinterpractice1.py	

Date	Team Member	Notes
10/04/2023	Bryan	Started by creating a lockercheck.py file with the goal of creating functions to check the status of lockers. I used a dictionary to hold key: value pairs of lockerNo and true or false. Used pickle to be able to save out the dictionary in case of power failure.
11/04/2023	Bryan	Created a function load the pickle file and check through the values until it found a false. It then converts the false to true and breaks.
12/04/202	Bryan	Created lockertester.py file as a sort of driver file to call the functions. Tested good, I could get results in terminal of the statues of all dictionary entries, could also reset the dictionary state to original all false.
17/04/2023	Bryan	Implementing tkinter. Progress moved to SCREEN OUTPUT section. Only major changes need to be added to this section.

Firestore Subsystem

FIREBASE DATA COLLECTION	
Files Used	
Lb_firestore.py	
food-for-dirt-testbed-firebase-adminsdk-jqk5x-108540edd9.json	

Date	Team Member	Notes
11/04/2023	Bryan	Very dodgy 1 st attempt to add a record to database. It was successful but added the data wrong. It connects to our firestore and creates a new collection with auto id document and does populate records. But this is wrong. Need to refine how, what and where we store data. BUILD A BETTER DB SCHEMA.
19/06/2023	Bryan	Created Lb_firestore.py Added a function that when called creates a record in firestore of: The locker used, the time of day, the date, and the userid that made the deposit. Tested successful.

Screen Output Display Subsystem

SCREEN OUTPUT / DISPLAY	
Files Used	
Lb_core.py	Empty.png
Lb_operational.py	Full.png
Lb_op_commands.py	Avail.png
Lb_maintenance.py	Lb_hold_screen.png
Lb_maintenance_commands.py	Lockers.ico
Lb_assets.py	Lockertest.dat
	Empty.png

Date	Team Member	Notes
		LOCKER CHECK
16/04/2023	Bryan	I started by creating tkinterpractice.py and set up a basic screen layout with an app name header, a body section for the changing screens, and a footer section with buttons for testing.
17/04/2023	Bryan	Added the original terminal logic functions to the tkinter files and tried to setup the buttons to call the functions. It worked and produced the same result but still in terminal. I added frames to the body section, 1 for each locker. Can get the frame name to change from empty to full. Semi working. Attempted to get frame image to change between empty.png and full.png failed.
19/04/2023	Bryan	Added in resizer function to change the images (tkinter is a lil bitch and does not like images) Added the current_state function to be called to display images determined by the lockers.dat file. Finally, a success the images change. Added the reset function to the buttons, tested. Resets the lockers.dat file to original state.
		QR READER
27/04/2023	Bryan	Setup QR read logic function to display in the body section of the tkinter screen. Tkinter is not only a lil bitch when it comes to images but is an absolute nightmare when it comes to video. I hate u tkinter.
28/04/2023	Bryan	Finally got tkinter to display the video. Because tkinter requires a loop to run and so does the camera read I had to take the image from the camera frame by frame and add and replace the image in the tkinter display. Display and read successfully, output QR decode still in terminal.

		REMODEL TO CLASSES
29/04/2023	Bryan	Rebuilding all functions and models into a class-based model. I started by creating the lb_core.py file, the absolute root of the tkinter app. I set up the core class as a parent to all other widgets. Created assets.py file to hold all constants and imported it to the lb_core.py file. Created lb_operational.py with the class lbOperational to be a child of core and display: header (app name), body to display all screens, and a footer for testing buttons. Created lb_op_Commands.py to hold all the logic and functions to be called by lbOperational. Created lb_maintenance.py with the class lbMaintenance to be a child of core and display: header (app name), body to display all screens, and a footer for testing buttons. Created lb_maintenance_Commands.py to hold all the logic and functions to be called by lbMaintenance.
30/04/2023	Bryan	Created hold_img function to display when app is first started
02/05/2023	Bryan	Created the open_cam, update_video, delete_img, close_cam functions in LB_op_commands.py to be called to control and read the video images. Created update_video_img function in lbOperational class to update the image displayed in the body frame. Added start cam and close cam button functionality. Tested successfully. Displays and reads QR code (outputs to terminal) close cam reverts to hold image.
05/05/2023	Bryan	Added locker logic locker_select and locker_state functions to lb_op_commands images will not display. After many attempts, I fed the problem into Microsoft's AI (Artificial Intelligence) and asked it for help. Spat out the answer almost immediately then informed me why mine was failing. Because I was calling the image inside a function it was getting garbage collected before adding it to the body display. Ugh, why are we bothering to finish this IT course? Altered the code to suit, images are now displaying. added a locker_reset function to lb_maintenance_commands.py to reset the lockertest.dat to an all "empty" value state.

06/05/2023	Bryan	Rewrote the locker_reset function. I was originally hard coding 6 lockers into a dictionary. Altered to take a value from assests.py of lockerTotal and created a loop to build the dictionary from.
07/05/2023	Bryan	Rewrote the locker_state function. It was originally set to display 6 hard coded frames and change the image. Altered to create a frame for each key taken from the lockertest.dat file and create a frame, then assigns an image as it did before based on the value. Using a loop to build out a grid system. **Need to alter the loop by rows and columns in case of large numbers of lockers. Tested successfully.
14/05/2023	Bryan	Added Menu.py Attempted to set up a way to change things like lockerTotal. Menu_bar is functional. Creating functions for settings is very in depth probably remove!

Design and Style Constants

Colors

- | | | |
|-------------------|-------------------|-------------------|
| • Primary color | green - #547817 | bpdPrimaryColor |
| • Secondary color | black - #272727 | bpdSecondaryColor |
| • Accent color | Blue - #0018FF | bpdPAccentColor |
| • Card background | White - #FFF7F6F1 | bpdCardBgColor |
| • White color | White - #ffffff | bpdWhiteColor |
| • Dark color | Black - #000000 | bpdDarkColor |

Images (store in discord design stuff)

- Splash screen top icon – simple green colored blob
- Splash screen image – simple line drawing cartoon of person putting food waste in a bin.
- Welcome screen image – something to represent what the app is for

All used images are found in: assets/images/

Image sizes

- Should not matter, high quality images at small scale are what we need.

Text styling

- | | | | |
|-------------------|------------|----------------|--------------------|
| • headlineMedium: | montserrat | fontSize: 24.0 | FontWeight: w700 |
| • headlineSmall: | poppins | fontSize: 24.0 | fontWeight: w700 |
| • displayLarge: | poppins | fontSize: 16.0 | fontWeight: w600 |
| • displaySmall: | poppins | fontSize: 14.0 | fontWeight: w600 |
| • bodyLarge: | poppins | fontSize: 16.0 | fontWeight: normal |
| • bodyMedium: | poppins | fontSize: 14.0 | fontWeight: normal |

Text

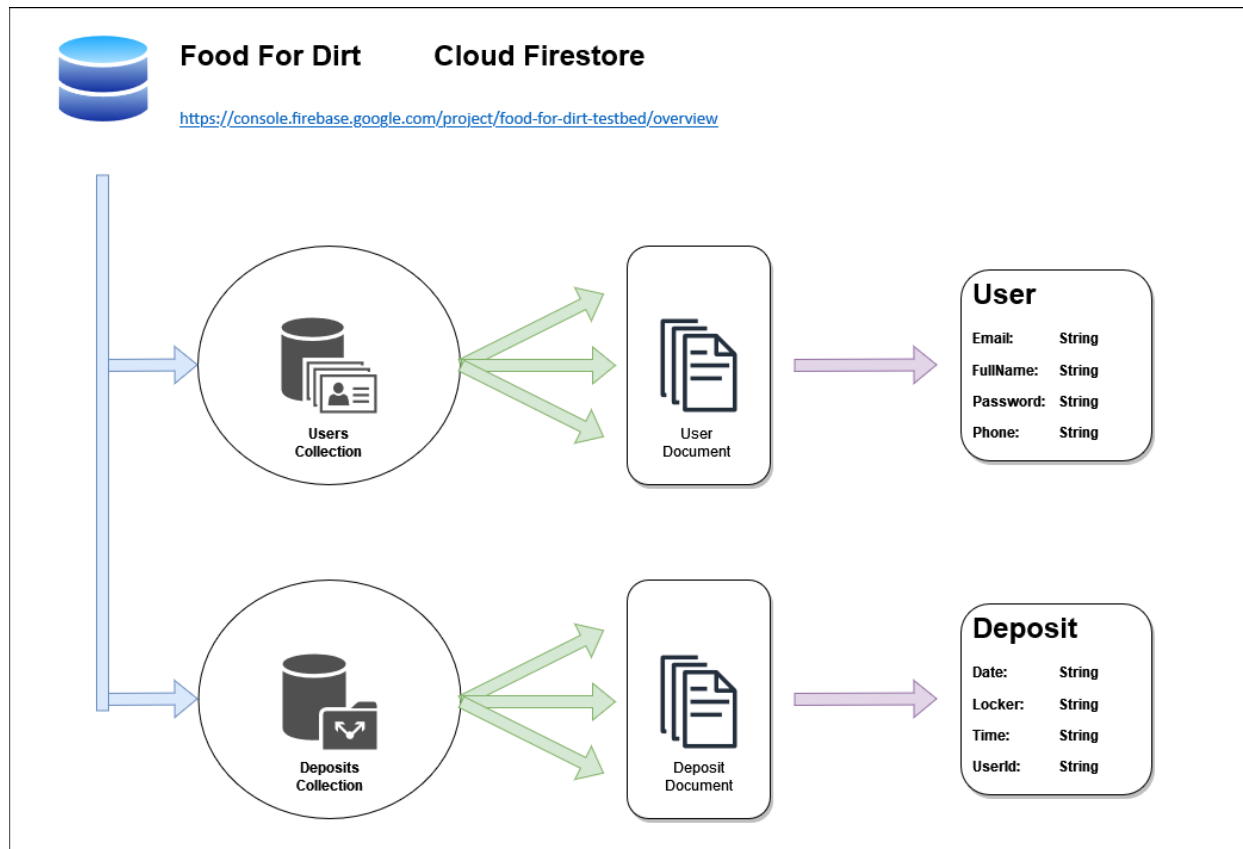
- | | |
|--------------------|--------------------------------|
| • App name | - Food For Dirt |
| • App tagline | - Home Eco Bins for Apartments |
| • Welcome title | - Welcome |
| • Welcome Subtitle | - Bin it |
| • Login tile | - Welcome back |
| • Login subtitle | - don't be shy :) |
| • Sign up title | - Create an Account |
| • Sign up subtitle | - Signup now! |

More added during construction of app. A full listing can be found in:

Lib/src/constants/text_strings.dart

Database Schema

<https://console.firebase.google.com/project/food-for-dirt-testbed/overview>



Firebase Authentication

- Email (required)
- Password (required)

User Collection – each user will have their own document created on signup via user application.

- FullName
- Email
- Phone Number
- Password (will be removed for security)

Deposits collection – each use of locker system creates a document containing.

- Date
- Time
- Locker Used
- User Id

Meeting Records

MEETING 1

	Project title:	SMART LOCKER SYSTEM		
	Facilitator:	A/Prof Thinh Nguyen		
	Team Members:	Bryan France, Peter Hans, Darcy Williams		
	Date:	29/01/23		
	Time:	8pm		
	Location	Discord		
Meeting Objective(s):				
Initial conversation regarding project, determine who is doing what and spitball possible issues both on the project itself and us working around				
Attendance:				
Present		Absent		
Bryan France Peter Hans Darcy Williams				
Agenda/Issues:				
Topic 1	Distributing workload for project proposal			
Topic 2	Discussing questions to ask project sponsor			
Topic 3	Creating the framework of the sequence diagram			
Next Meeting:				
Date:	29/01/23	Time:	8pm	Location: Discord
Objective(s):	First check in with work progress, re-assess workload, distribute remaining work			

MEETING 2

	Project title:	SMART LOCKER SYSTEM		
	Facilitator:	A/Prof Thinh Nguyen		
	Team Members:	Bryan France, Peter Hans, Darcy Williams		
	Date:	06/03/2023		
	Time:	8pm		
	Location	Discord		
Meeting Objective(s):				
Project Plan discussion				
Figure out what has done and what is left to do for the project Plan				
Attendance:				
Present		Absent		
Bryan France Peter Hans Darcy Williams				
Agenda/Issues:				
Topic 1	Use Gantt to figure out what is left and how In Progress tasks are going.			
Topic 2	Team availability over the next week and a half to work on tasks			
Topic 3	Discuss long term strategies. What is the roll out over the next few months? Create a new Gantt for project development lifespan. Figure out how we can all work on the same code base (maybe set up a GitHub).			
Topic 4	Organize catch up with Assefa and or Thinh Figure out best day when we can all make it. Reminder to continue to add any questions for Assefa and Thinh to doc			
Next Meeting:				
Date:	06/03/2023	Time:	8pm	Location: Discord
Objective(s):	Discuss project start			

MEETING 3

 VICTORIA UNIVERSITY MELBOURNE AUSTRALIA	Project title:		SMART LOCKER SYSTEM		
	Facilitator:		A/Prof Thinh Nguyen		
	Team Members:		Bryan France, Peter Hans, Darcy Williams		
	Date:		25/03/2023		
	Time:		8pm		
	Location		Discord		
Meeting Objective(s):					
Discuss project start - Objectives and Goals Figure out where we are at and what has been done (research)					
Attendance:					
Present		Absent			
Bryan France Peter Hans Darcy Williams					
Agenda/Issues:					
Topic 1	Find out what the team has researched and how much practice they have had with IDEs (Integrated Development Environment) and languages, also GitHub				
Topic 2	Programming language chosen - will be written in kotlin for android, swift for IOS, using Flutter and Android studio to create. Connecting to Firebase. Figure out how to run Xcode on PC or acquire a mac.				
Topic 3	Discuss progress report documentation and how we will add to it. Discuss design and what is needed. Discuss app directory structure, and naming convention “bpdSomethin”				
Topic 4	Figure out what each of us is doing and will be focusing on (ref. GANTT)				
Next Meeting:					
Date:	25/03/2023	Time:	8pm	Location:	Discord
Objective(s):	Finalization of login app, organize meeting with Thinh and Assefa				

MEETING 4

	Project title:	SMART LOCKER SYSTEM			
	Facilitator:	A/Prof Thinh Nguyen			
	Team Members:	Bryan France, Peter Hans, Darcy Williams			
	Date:	11/04/23			
	Time:	8pm			
	Location	Discord			
Meeting Objective(s):					
Discuss fine tuning / polishing of app. Discuss locker system project start.					
Attendance:					
Present		Absent			
Bryan France Peter Hans Darcy Williams					
Agenda/Issues:					
Topic 1	What still needs to be done to the app? clean code, polish screens, extras? What do we want the QR image data to be when scanned? user ID? Design / Styling of app?				
Topic 2	Locker system - find out how much research the team has done. QR readers / code to communicate with the firestore database.				
Topic 3	Review the Gantt figure out where we are at and what needs to be done in future?				
Topic 4	Questions for Thinh and Assefa? make a list				
Next Meeting:					
Date:	11/04/23	Time:	8pm	Location:	Discord
Objective(s):	Discuss locker bank system start				

MEETING 5

	Project title:	SMART LOCKER SYSTEM			
	Facilitator:	A/Prof Thinh Nguyen			
	Team Members:	Bryan France, Peter Hans, Darcy Williams			
	Date:	23 / 04 / 23			
	Time:	8pm			
	Location	Discord			
Meeting Objective(s):					
Discuss locker bank system start					
Attendance:					
Present		Absent			
Bryan France Peter Hans Darcy Williams					
Agenda/Issues:					
Topic 1	Find out what research the team has done?				
Topic 2	What part of the system do the team members feel confident to work on?				
Topic 3	Discuss locker bank system timeline				
Next Meeting:					
Date:	23 / 04 / 23	Time:	8pm	Location:	Discord
Objective(s):	Catch up				

MEETING 6

	Project title:	SMART LOCKER SYSTEM		
	Facilitator:	A/Prof Thinh Nguyen		
	Team Members:	Bryan France, Peter Hans, Darcy Williams		
	Date:	28/05/2023		
	Time:	8pm		
	Location	Discord		
Meeting Objective(s):				
Catch up. Progress meeting				
Attendance:				
Present		Absent		
Bryan France Peter Hans Darcy Williams				
Agenda/Issues:				
Topic 1	Discuss the TO DO List doc created.			
Topic 2	Work out who wants to do what on the list			
Topic 3	Start doing it!!			
Next Meeting:				
Date:	28/05/2023	Time:	8pm	Location: Discord
Objective(s):	Figure out what is done / needs to be done			

MEETING 7

 VICTORIA UNIVERSITY MELBOURNE AUSTRALIA	Project title:	SMART LOCKER SYSTEM		
	Facilitator:	A/Prof Thinh Nguyen		
	Team Members:	Bryan France, Peter Hans, Darcy Williams		
	Date:	25/06/2023		
	Time:	8pm		
	Location	Discord		
Meeting Objective(s):				
Catch up. Progress meeting				
Attendance:				
Present		Absent		
Bryan France Peter Hans		Darcy Williams		
Agenda/Issues:				
Topic 1	Discuss the TO DO List. find out what has been done and needs to be done			
Topic 2	Security. Need to get this done before the 11th to allow time for testing			
Topic 3	Progress report discuss what still needs to be added to the report. Database schema needs to set			
Topic 4	Need to get app onto phones / tablets and do live tests. Make at least 1 in person time to meet that suit all. Try to get Assefa and Thinh to attend			
Next Meeting:				
Date:	25/06/2023	Time:	8pm	Location: Discord
Objective(s):	Last meeting			

MEETING 8

	Project title:	SMART LOCKER SYSTEM		
	Facilitator:	A/Prof Thinh Nguyen		
	Team Members:	Bryan France, Peter Hans, Darcy Williams		
	Date:	26/06/2023		
	Time:	8pm		
	Location	Discord		
Meeting Objective(s):				
FINAL MEETING. Figure out what we have not done yet				
Attendance:				
Present		Absent		
Bryan France Peter Hans		Darcy Williams		
Agenda/Issues:				
Topic 1	Discuss project and what is due before applied project 2			
Topic 2	Discuss what will be require during applied project 2			
Topic 3	Figure out when applied project 2 starts			
Next Meeting:				
Date:	----	Time:	----	Location: ---
Objective(s):	----			